

Knee extension and flexion muscle power after anterior cruciate ligament reconstruction with patellar tendon graft or hamstring tendons graft: a cross-sectional comparison 3 years post surgery

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Abstract Hamstring muscles play a major role in knee-joint stabilization after anterior cruciate ligament (ACL) injury. Weakness of the knee extensors after ACL reconstruction with patellar tendon (PT) graft, and in the knee flexors after reconstruction with hamstring tendons (HT) graft has been observed up to 2 years post surgery, but not later. In these studies, isokinetic muscle torque was used. However, muscle power has been suggested to be a more sensitive and sport-specific measures of strength. The aim was to study quadriceps and hamstring muscle power in patients with ACL injury treated with surgical reconstruction with PT or HT grafts at a mean of 3 years after surgery. Twenty subjects with PT and 16 subjects with HT grafts (mean age at follow up 30 years, range 20–39, 25% women), who were all included in a prospective study and followed the same goal-based rehabilitation protocol for at least 4 months, were assessed with reliable, valid, and responsive tests of quadriceps and hamstring muscle power at 3 years (SD 0.9, range 2–5) after surgery. The mean difference between legs (injured minus uninjured), the hamstring to quadriceps (H:Q, hamstring divided by quadriceps) ratio, and the limb symmetry index (LSI,

injured leg divided by uninjured and multiplied by 100) value, were used for comparisons between the groups (analysis of variance). The mean difference between the injured and uninjured legs was greater in the HT than in the PT group for knee flexion power (–21.3 vs. 7.7 W, $p = 0.001$). Patients with HT graft had lower H:Q ratio in the injured leg than the patients with PT graft (0.63 vs. 0.77, $p = 0.012$). They also had lower LSI for knee flexion power than those in the PT group (88 vs. 106%, $p < 0.001$). No differences were found between the groups for knee extension power. The lower hamstring muscle power, and the lower hamstring to quadriceps ratio in the HT graft group than in the PT graft group 3 years (range 2–5) after ACL reconstruction, reflect imbalance of knee muscles after reconstruction with HT graft that may have a negative effect on dynamic knee-joint stabilization.

Keywords Anterior cruciate ligament ·
Reconstructive surgical procedures · Rehabilitation ·
Muscle strength

Introduction

The anterior cruciate ligament (ACL) is the primary restraint to anterior translation of the tibia relative to the femur, and the hamstring muscles play a major role in protecting the ACL and compensating for the loss of stability in the ACL-deficient knee. Weak hamstring muscles and low co-activity in relation to quadriceps are associated with a risk of ACL injury [9], and hamstring weakness is associated with poorer knee function after ACL injury [17, 30].

The patellar tendon (PT) graft has been the gold standard in ACL reconstruction, but during the last decade, the use of hamstring tendons (HT) graft has increased [10].

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Systematic reviews show that there is no evidence to indicate that one graft type is better than the other regarding the knee stability, pain, range of motion, activity level or patient-reported function after surgical reconstruction [4, 10, 27].

Muscle strength tests are commonly used in the evaluation of recovery after ACL reconstruction and subsequent training [23]. Weakness of the knee extensor muscles after reconstruction with PT graft, and in the knee flexor muscles after reconstruction with HT graft has been observed up to 24 months post surgery according to a literature review [5]. Based on the results of this review, it was suggested that there is no difference in recovery of muscle strength between PT and HT grafts groups more than 2 years after surgery [5]. In these studies, isokinetic muscle torque was tested. However, the ability to produce high force quickly is important for sports and injury protection. Therefore, muscle power has been suggested to be more sport-specific for not only force but also velocity is measured [16].

The aim was to study quadriceps and hamstring muscle power in patients with ACL injury treated with surgical reconstruction with PT graft or HT graft at a mean of 3 years after surgery.

Subjects and methods

Subjects

Twenty subjects with PT graft and 16 subjects with HT graft (mean age at follow up 30 years, range 20–39, 25% women), were included in this cross-sectional, case–control study. They were all from a cohort of 121 patients with ACL injury included in a randomized controlled trial on training and surgical reconstruction versus training only (the KANON study, ISRCTN84752559) [7]. The outcome of the RCT will be presented separately. Patients with a complete ACL rupture less than 4 week old, aged 18–35 years, on a moderate to high physical activity level (corresponding to 5–9 on the Tegner Activity Scale), were included in the RCT. Inclusion and exclusion criteria in the RCT have been described [7]. All patients followed a moderately aggressive training programme under the supervision of physical therapists. The physical therapists had all accepted to follow the same goal-based training protocol. The training protocol included four levels of function. Each level contained examples of exercises (outlined by the participating physical therapists) and ended with a set of goals for range of motion, muscle function, and functional performance (activities of daily living and more strenuous activities). Time intervals for

each level were suggested, but these were secondary to the goals. When the goals of one level were reached, progression to the next level was made. Pain, swelling, and general discomfort slowed the progression. The supervised training continued for at least 4 months until the final goals were reached. The Research Ethics committee of Lund University approved the study, and all subjects gave their written informed consent.

Four experienced senior knee surgeons (7–17 years of experience as orthopaedic specialists) (surgeons A, B, C, and D) performed the ACL reconstructions. The surgeons performed annually a mean of 62 (range 42–80) ACL reconstructions. They worked in orthopaedic units with close connection and collaboration, and they employed the same principles for ACL reconstruction. In the majority of the cases, the graft choice was based on the surgeon's preference. All patients were examined under anaesthesia, diagnostic arthroscopy was performed and, if necessary, meniscal surgery was carried out prior to the ACL reconstruction. The graft was tightened and fixated in 5°–20° of knee flexion. A notch plasty was made if there was a risk of interference in the intercondylar notch. In the PT procedures, the femoral tunnel was drilled through the anteromedial portal. The central third of the ipsilateral PT with bone blocks was used. The bone blocks were fixated in the tunnels on the femoral and tibial side with standard round head of 7 × 25 mm titanium cannulated interference screws. The graft was harvested through a two incision technique. In the hamstring procedures, where a quadrupled semitendinosus and gracilis tendon graft was used, the femoral tunnel was drilled through the tibial tunnel, and a drill guide was used. A Transfix (Artrix, Naples, FL) on the femur was used in all but two cases. In these two cases, a Rigid Fix (Depuy Mitek, Johnson & Johnson, Gateway) was used. The tibial fixation was performed with soft threaded interference screws in all cases. The number of reconstructions with PT and HT grafts for the four surgeons was: surgeon A: 15 PT and 0 HT, B: 1 PT and 11 HT, C: 4 PT and 4 HT, and D: 0 PT and 1 HT.

Muscle function data for the present study was collected from a subset of the RCT at a separate occasion, not a part of the RCT protocol, at a mean of 3 years (SD 0.9, range 2–5 years) after the injury [2]. Thirty-six of the subjects (nine women) included in the assessment of muscle function had been treated with surgical reconstruction and training. Twenty of these were reconstructed with PT graft (five women) and 16 with HT graft (four women). Patient characteristics, including self-reported outcomes assessed by the knee injury and osteoarthritis outcome score (KOOS) [24, 25], are given in Table 1. One patient had postoperative arthrofibrosis in the reconstructed knee (PT graft).

Table 1 Characteristics of the patients

Characteristic	Patellar tendon graft group (n = 20)	Hamstring tendons graft group (n = 16)
Age (years) ^a	32 (5.0)	28 (4.4)
Women, n (%)	5 (25%)	4 (25%)
BMI ^a	25.2 (4.2)	23.6 (2.5)
Injured knee, right, n (%)	11 (55)	8 (50)
Associated meniscal tears at baseline, n (%)	13 (65)	7 (44)
>2-year-old contralateral ACL injury, n (%)	2 (10)	0
>2-year-old contralateral ACL reconstruction, n (%)	4 (20)	3 (19)
Tegner activity level ^b	3 (1–6)	4 (2–7)
KOOS subscales ^a		
Pain	88 (11.3)	88 (13.9)
Symptoms	80 (16.0)	79 (19.5)
ADL	94 (9.2)	95 (9.6)
Sport/Rec	69 (26.3)	77 (22.1)
QOL	69 (26.7)	65 (27.7)

BMI body mass index

^a Mean (SD)

^b Median (quartiles)

Assessment

The subjects were told not to participate in any strenuous activity the day before the test session. They were also asked to continue their activities of daily living as usual before the test session. Prior to testing, the subjects completed a warm-up consisting of 5 min of stationary cycling, two times ten squats, two times ten toe rises, ten jumps on both legs, and five jumps on the right and left legs, respectively. Knee extension power test and knee flexion power test were performed in a randomized order. The right leg was tested first. The subjects wore shorts, T-shirt, and sneakers. Tubigrip[®] stockings (MEDLOCK Medical, Oldham, UK) covered both knees to disguise scars from reconstructive surgery, i.e. the test leader was blinded regarding injured knee.

The muscle strength tests were performed as described by Neeter et al. [19]. The muscle strength tests were supervised by the same test leader, and standardised verbal instructions and encouragement were given. Before each strength test, the subjects completed ten repetitions at a sub-maximum weight, followed by five repetitions using a somewhat higher sub-maximum weight. The subjects also performed practice trials before the measurements were taken (pre-test). If a subject reported pain in the leg during the test, which occurred in a few cases,

this particular test was cancelled and the measurement was repeated. The strength tests were chosen to reflect quadriceps and hamstring muscle power during knee extension and knee flexion, respectively (open kinetic chain exercises).

Knee extension power test

This test was performed in a knee extension weight training machine (Precor, Competition Line, Borås, Sweden). The subject was seated using an individual seating position. The test was performed one leg at a time, and the other leg was fixed in place using a strap. Prior to testing, an appropriate starting weight was selected based on the subject's performance during the pre-test, which consisted of a single sub-maximum repetition. On a given signal, the subject was asked to extend his/her knee as quickly and forcefully as possible from approximately 110° of knee flexion to full knee extension (0° of flexion). The distance the weight stack was lifted and the time it took to fully extend the knee was measured with a linear encoder connected to the weight stack of the machine. The average power was calculated by Muscle Lab, a computerised muscle function measuring system (Ergotest Technology, Oslo, Norway). In all, the subjects performed five maximum trials at five weight levels. The weight was increased by 5 kg for each trial. The rest period between trials was 30 s, which was considered to be sufficient for full recovery. After the five trials, the Muscle Lab calculated the maximum average power, which was later compared with the results from the other leg.

Knee flexion power test

The knee flexion power test was performed in a knee flexion weight training machine (Precor, Competition Line). The subject was seated using an individual seating position. The same procedure was used as that for the knee extension power test. The test was performed from full knee extension (0° of flexion) to approximately 110° of knee flexion.

Statistical analysis

The absolute values, the hamstring to quadriceps (H:Q) ratio (hamstring divided by the quadriceps), and the limb symmetry index (LSI) value (calculated by dividing the result for the injured leg by that of the uninjured leg and multiplying by 100), were used for comparisons between the groups. An LSI $\geq 90\%$ for an individual in a single test was considered normal [19]. Analysis of variance, adjusting for gender, was used for the between-group comparisons. Spearman's rank correlation coefficient (r_s)

was used to assess the association between muscle power and patient-reported function. A level of $p \leq 0.05$ was chosen to indicate statistical significance.

Results

The patients with HT graft had lower values for hamstring muscle power and lower H:Q ratio of their injured leg, and lower LSI for knee flexion, than the patients with PT graft. No differences were found between the groups for knee extension power.

Absolute values of knee extension and flexion power

The subjects in the HT graft group had 15% lower hamstring muscle power in the injured leg than the subjects in the PT graft group (153.0 W vs. 179.1 W, $p = 0.085$) (Table 2). No differences were observed between the groups for knee flexion power of the uninjured leg or for knee extension power of the injured or uninjured legs (relative difference of 3% between groups) (Table 2). To reduce the effect of confounding factors, since the study was non-randomized, the absolute difference between the injured and uninjured legs was also analysed. The mean difference between the injured and uninjured legs (injured minus uninjured) was greater in the HT group than in the PT group for knee flexion ($p = 0.001$), but not for knee extension ($p = 0.919$) (Fig. 1). There were too few women to permit comparisons of possible gender-related differences. However, in the HT group, all women had lower knee flexion power than the men. This was not observed in the PT group (Fig. 2).

Hamstring to quadriceps ratio

The patients with HT graft had lower H:Q ratio in the injured leg than the patients with PT graft (0.63 vs. 0.77, $p = 0.012$), but not in the uninjured leg (0.67 vs. 0.70, $p = 0.632$) (Fig. 3).

Limb symmetry index

The patients with HT graft had lower LSI for knee flexion power than the patients with PT graft (88 vs. 106%, $p < 0.001$). Both groups had a mean LSI of 94% for knee extension power (Table 2). Sixteen (84%) of the subjects in the PT graft group and 8 (50%) of the subjects in the HT graft group had normal LSI (i.e. $\geq 90\%$) for knee flexion power ($p = 0.065$). Thirteen (65%) of the subjects in the PT graft group and 12 (75%) of the subjects in HT graft group had normal LSI for knee extension power ($p = 0.718$).

Table 2 Mean and standard error (SE) for the knee extension and flexion muscle power tests in the injured (inj) and uninjured (uninj) legs, limb symmetry index (LSI), and mean difference (95% CI) between the patellar tendon graft and hamstring tendons graft groups (analysis of variance, adjusted for gender)

	Patellar tendon graft group ($n = 20$) ^a			Hamstring tendons graft group ($n = 16$) ^a			Patellar tendon graft versus hamstring tendons graft ^b		
	Inj leg	Uninj leg	LSI	Inj leg	Uninj leg	LSI	Inj leg	Uninj leg	LSI
Knee extension (W)	241.2 (15.2)	255.2 (12.7)	94 (3.4)	248.2 (17.0)	263.6 (14.2)	94 (3.8)	-7.0 (-53.4 to 39.5)	-8.4 (-46.9 to 30.3)	0 (-9.9 to 10.6)
Knee flexion (W) ^c	179.1 (9.9)	171.4 (10.0)	106 (2.9)	153.0 (10.8)	174.3 (10.9)	88 (3.2)	26.1 (-3.8 to 56.0)	-2.9 (-32.9 to 27.2)	18 (9.1 to 26.7)

^a Values are expressed as mean (SE)

^b Values are expressed as mean difference (95% CI)

^c $n = 35$; one patient in the patellar tendon graft group did not perform the knee flexion power test

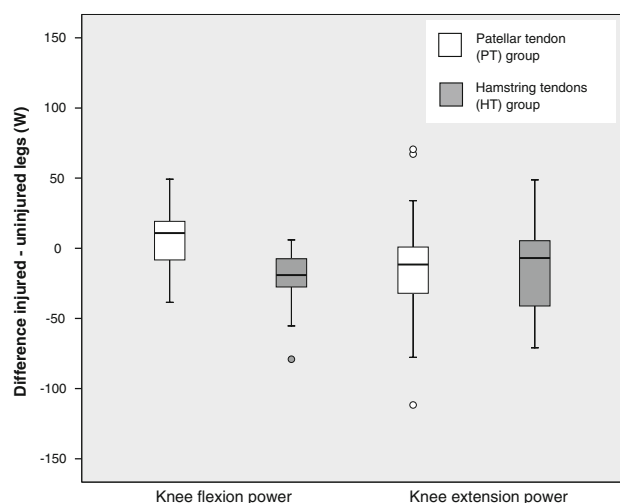


Fig. 1 Absolute difference between the injured and uninjured legs (injured–uninjured) for knee flexion and extension power in the patellar tendon (PT) and hamstring tendons (HT) grafts groups. The mean difference between the injured and uninjured legs was greater in the HT group than in the PT group for knee flexion ($p = 0.001$), but not for knee extension ($p = 0.919$). The box includes the first to third quartiles with median values shown as a line through the box. The whiskers show values below 1.5 box lengths from the upper or lower edge of the box. The circles denote outliers (i.e. the value was between 1.5 and 3 box lengths from the upper edge of the box)

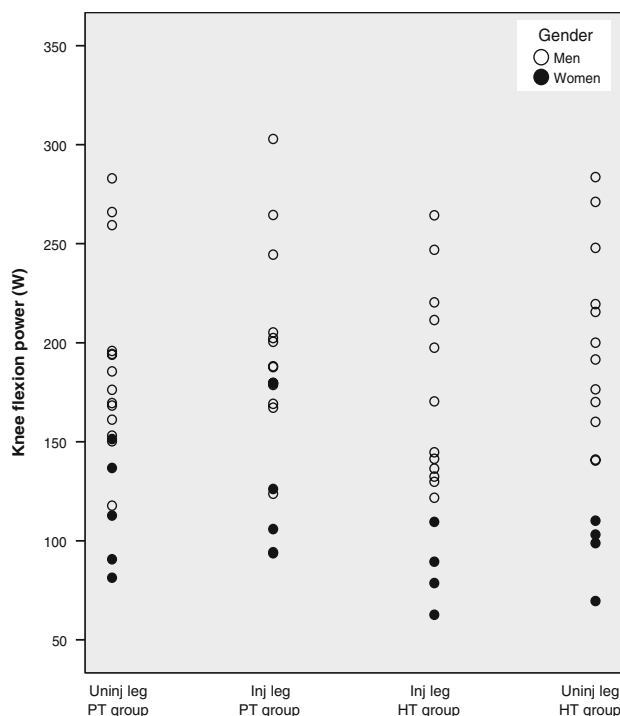


Fig. 2 Knee flexion power (W) for the men and the women in the injured (inj) and uninjured (uninj) legs in the patellar tendon (PT) and hamstring tendons (HT) grafts groups

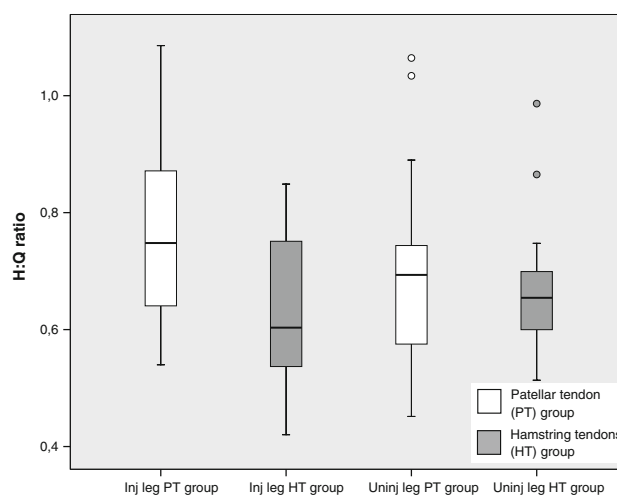


Fig. 3 Hamstring to quadriceps (H:Q, hamstring divided by quadriceps) ratio in the injured (inj) and uninjured (uninj) legs in the patellar tendon (PT) and the hamstring tendons (HT) grafts groups. The patients with HT graft had lower H:Q ratio in the injured leg than the patients with PT graft ($p = 0.012$), but not in the uninjured leg ($p = 0.632$). The box includes the first to third quartiles with median values shown as a line through the box. The whiskers show values below 1.5 box lengths from the upper or lower edge of the box. The circles denote outliers (i.e. the value was between 1.5 and 3 box lengths from the upper edge of the box)

Association between muscle power and patient-reported function

To study the clinical relevance of muscle imbalance, self-reported function, assessed by the KOOS subscales sport and recreation function, and quality of life, respectively, were correlated with the H:Q ratio of the injured leg and the LSI for knee flexion and extension power. KOOS QOL correlated with LSI for knee extension power in the PT group ($r_s = 0.49$, $p = 0.027$), and with LSI for knee flexion power ($r_s = 0.61$, $p = 0.011$) and H:Q ratio ($r_s = 0.46$, $p = 0.070$) in the HT group. This indicates that lower degrees of asymmetry, i.e. a smaller difference between hamstring and quadriceps muscle strength (higher H:Q ratio) and a smaller side-to-side difference (higher LSI) are associated with better knee-related quality of life, including confidence in the knee. There was no or low correlation between KOOS sport/recreation and H:Q ratio and LSI values in the PT and HT groups ($r_s = -0.2$ – 0.3), indicating that H:Q ratio and LSI are not associated with self-reported difficulty with functions, such as kneeling, squatting, running, twisting and jumping.

Discussion

The principal findings of this study were that 3 years (range 2–5) after reconstructive surgery, the patients with

HT graft had lower hamstring muscle power and lower hamstring to quadriceps ratio in their injured leg than the patients with PT graft. There were no differences between the groups in quadriceps muscle power. To our knowledge, our study is the first to compare the knee muscle power in patients with ACL reconstruction with PT or HT grafts.

A limitation of the study is that the choice of graft was not controlled. The purpose of the main RCT was to compare surgical with non-surgical treatment [7], and accordingly the graft choice was not randomized. In the majority of the cases, the graft choice was based on the individual surgeon's preference of surgical method, rather than on patient-related factors, such as gender, activity level, or occupation. Thus, we suggest that the overall risk of a selection bias directing the individual patient to one of the methods was small. Two of the surgeons (A and B) performed 75% of the reconstructions and those two had different preferences of graft choice. However, they worked at the same hospital and had developed their methods and surgical skills in reconstructing the ACL at the hospital together. Both were experienced surgeons, decreasing the risk that a difference in technical skills between the surgeons could have influenced the results. Surgeon C performed eight reconstructions, with an equal number of PT and HT grafts. The graft choice in these cases was more likely based on the surgeon's own experience that some patients were more suited for one of the methods. For example, soccer players were, according to his opinion, more suitable for PT graft due to the bone-to-bone healing and patients with kneeling activities or patello-femoral pain preferably had a reconstruction with HT graft. Thus in these eight cases there is a likely selection bias with regard to patient and surgical method. Despite these limitations, we believe that a fair comparison regarding muscle strength between the patients operated on with different grafts can be made. The patients in the HT group were younger (28 vs. 32 years), had lower BMI (23.6 vs. 25.2), and higher KOOS Sport/Rec (77 vs. 69) than those in the PT group (Table 1). These confounding factors of potential interest, i.e. age, BMI, and self-reported outcomes, are all related to the individual and not to the operated leg only. Thus, to reduce the influence of these possible confounding factors, we used relative values comparing muscle power within the operated leg (H:Q ratio), and between the operated and the non-operated legs (absolute difference and LSI).

In absolute values of muscle power, we found that the subjects in the HT graft group had 15% lower hamstring muscle power in their injured leg than the subjects in the PT graft group. This difference was not statistically significant ($p = 0.085$), probably due to a too small sample size, but we considered it clinically relevant. The

corresponding difference between the groups in the uninjured leg for knee flexion power and in the injured and uninjured legs for knee extension power was only 3% (Table 2). In addition, the absolute mean difference between the injured and uninjured legs was greater in the HT group than in the PT group for knee flexion ($p = 0.001$), but not for knee extension ($p = 0.919$). Our findings of weaker hamstring muscles in the subjects with HT graft compared to those with PT graft, and no difference between graft groups in quadriceps strength, are similar to those of a randomized controlled study [3]. In this RCT, patients with HT graft had an 11% decrease in isokinetic knee flexion torque at 240° per second compared with those with PT graft at a 3-year follow-up, but there was no difference in isokinetic knee extension torque at the same velocity [3]. The authors found no differences between the graft groups in isokinetic knee extension or flexion torque at lower velocities (60 and 180° per second) [3], which is in line with other studies [14, 18]. Thus, more sport-specific measures of muscle strength, such as power and isokinetic muscle torque at high angular velocities, may be sensitive in detecting strength deficiency.

The hamstring muscles play a major role in preventing or decreasing anterior and rotatory displacement of the tibia relative to the femur. Systematic reviews show increased side-to-side sagittal laxity [4, 10, 27] and tibial rotation [28] after ACL reconstruction with PT and HT grafts, and it was suggested that the excessive tibial rotation will lead to osteoarthritis (OA) over time [28]. It was also reported that patients with HT graft (semitendinosus and gracilis tendons) were weaker in internal tibial rotation strength compared to patients with PT graft and to uninjured matched controls [29], possibly putting the reconstructed ACL at a risk of reinjury [26]. Because the hamstring muscles are ACL agonists, recovery of hamstring muscle strength is of importance after ACL reconstruction, and it may be argued that this is of particular importance after reconstruction with HT graft.

We found that in the HT graft group, all women had lower knee flexion power than men (Fig. 1). This gender-related difference was not found in the PT graft group. Although there were too few women in our study to permit gender-related comparisons, this finding is in line with that observed by Gobbi et al. [8]. It has also been reported that women with HT graft have poorer patient-reported function [15], poorer knee-joint stability [8, 13, 20], and higher failure rate (defined as abnormal laxity or revision surgery) than men [20].

The lower H:Q ratio in the HT graft group than in the PT graft group may reflect imbalance of knee muscles. Higher H:Q ratio has been reported in subjects with ACL reconstruction than in uninjured subjects [11], and it has been

suggested that higher H:Q ratio reduces the shear on the knee-joint [9] and is needed to protect the knee from further injury [11, 12]. We found that the patients in the HT group with higher H:Q ratio were those with better knee-related quality of life, including confidence in the knee. An association between weak hamstring muscles in relation to quadriceps and patient-reported function (Cincinnati and Lysholm knee scores) was reported by others [17, 30]. However, in contrast to the KOOS, knee-related quality of life is not included in the Cincinnati or Lysholm scores. Thus, based on the results of our study and previous ones, a high H:Q ratio may be an important factor for dynamic knee-joint stabilization and seems to be relevant to the patient after ACL reconstruction.

The lower LSI, i.e. a greater side-to-side difference, for knee flexion power in the HT group (injured leg 12% weaker than uninjured) than in the PT group (injured leg 6% stronger than uninjured), may reflect imbalance of knee flexor muscle strength between legs. More than 80% of the subjects in the PT group, but only 50% of the subjects in the HT group had normal LSI (i.e. $\geq 90\%$) for knee flexion power. There was no difference between groups in LSI for knee extension power (94% in both groups), or any obvious difference between the groups in number of subjects with normal LSI (65% in the PT group and 75% in the HT group). A high LSI, i.e. a small difference between the injured and uninjured legs, indicates a good muscle function, whereas imbalance between legs in muscle strength is associated with an increased risk of injury [9]. We found that the patients in the HT group with higher LSI for knee flexion were those with better knee-related quality of life. This association was also found in the patients with PT graft (higher LSI for knee extension was associated with better KOOS QOL score), although the correlation was lower in the PT group ($r_s = 0.49$) than in the HT group ($r_s = 0.61$). An association between a smaller side-to-side difference and better patient-reported function, assessed by the Lysholm knee score, was reported by others [18]. Balance in muscle strength between legs may also be of importance in the long term. Knee muscle strength, especially hamstring muscle strength, is a significant contributing factor to the one-leg hop test for distance [21]. An abnormal LSI (i.e. $<90\%$) of the one-leg hop test at a 1-year follow-up predicted radiographic OA at 10 years in patients with ACL reconstruction [22]. Thus, balance in muscle strength between legs seems to be relevant to the patient and important for knee-joint stabilization and future OA.

Current training programmes include exercises that facilitate compensatory functional joint stabilization, after both surgical and non-surgical treatment of the ACL [1, 23]. The aim of the training is to restore neuromuscular function to the pre-injury or to a modified level of

activity by obtaining compensatory functional stability. The principles of training apply to patients with ACL injury and reconstruction, regardless of PT or HT grafts, since the training aims to resemble conditions in daily life and more strenuous activities [1]. Thus, the goal of training is the same for patients treated without surgical reconstruction, and for patients surgically treated with PT graft or HT graft. Factors such as age, gender, previous and desired activity level, type of injury, severity of injury, and type of graft are taken into account for each individual patient during training. To achieve functional stability, exercises where several muscles in the lower extremity and trunk are activated in synergies, are included. Strength, coordination, balance, and proprioception are all included in the exercises. The training protocols, including that in our study, are goal based. The training protocol in our study included four levels of function, ending with a set of goals for range of motion, muscle function, and functional performance (activities of daily living and more strenuous activities). When the goals of one level were reached, progression to the next level was made. The supervised training continued for at least 4 months until the final goals were reached.

The shift in surgical practice from PT graft to HT graft has been made with little evidence [3, 9, 26], and there is discussion on whether the use of HT graft leads to impaired muscle function [6]. From the results of our study, the hamstring muscles, which are important ACL agonists, seem to be negatively affected by the HT graft technique, causing an imbalance of knee muscle strength within the injured leg and between legs. Thus, the ability of the hamstring muscles to protect the knee after ACL reconstruction with HT graft may be impaired, with a possible negative effect on dynamic knee-joint stabilization, and a possible higher risk of graft rupture.

Conclusion

The lower hamstring muscle power and the lower hamstring to quadriceps ratio in the HT graft group than in the PT graft group 3 years (range 2–5) after surgery, reflect imbalance of knee muscles after reconstruction with HT graft which may have a negative effect on dynamic knee-joint stabilization.

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